Introduction to
Computer Organisation and Architecture
UCCD1133

Chapter 3

Basic Concept of Logic

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Outline

- Combinational circuit building blocks
  - Decoder
  - Multiplexer
  - Adder
  - Subtractor

Combinational Circuits Building Blocks

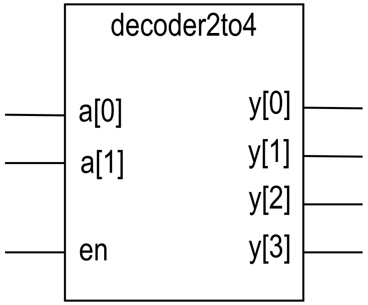
- Example of non-arithmetic building blocks:
  i. Decoder
  ii. Encoder and priority encoder
  iii. Demultiplexer
  iv. Multiplexer

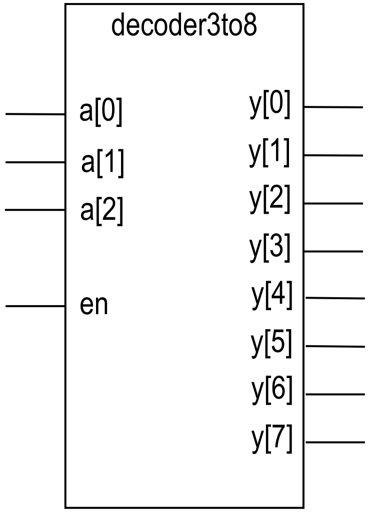
- Example of arithmetic building blocks:
  i. Adder
  ii. Subtractor
  iii. ALU (Arithmetic and Logic)
  iv. Multiplier
  v. Diviser
  vi. Comparator
  vii. Shifter

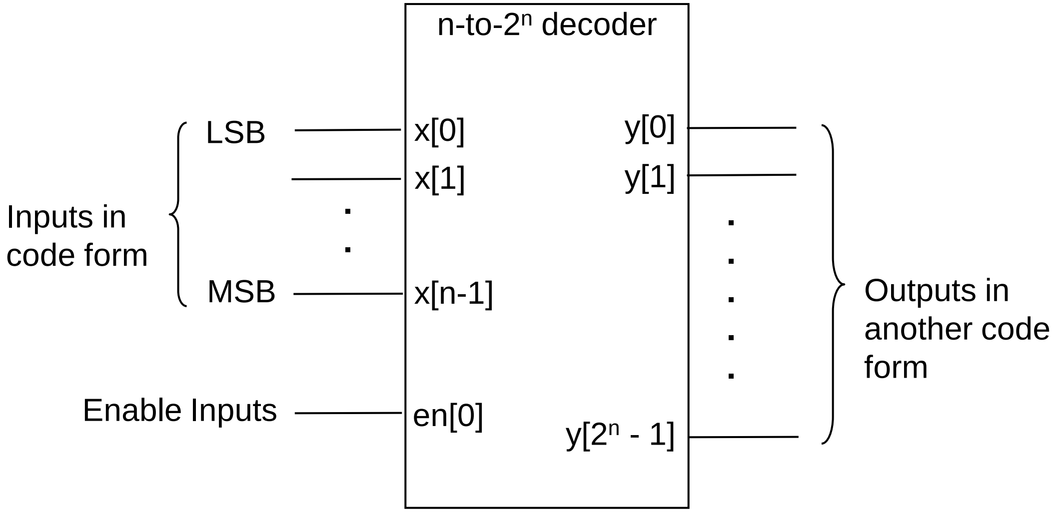
- What is standard circuits?
  i. Pre-designed circuits with at least correct functionalities
  ii. They speed up design work (modelling based on schematic, verification and implementation)
  iii. Collected as part of library

Decoder

- A decoder:
    i. Converts one type of code, x[n-1:0], into another type, y[2^n-1:0].
    ii. The enable inputs, en - to enable/ disable the outputs



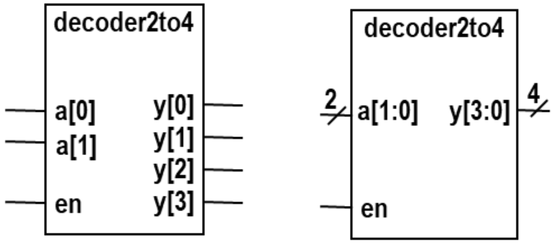




- A binary decoder
    i. Most basic type of decoder
    ii. Also known as n-to-2^n decoder – e.g. 3-to-8 decoder.
        ✓ Has n inputs and 2^n outputs.
    iii. Only one output (1-out-of-2^n) can be asserted
        ✓ To indicate/ identify the code that has been placed at input.
- Other types of decoder: display decoder

Decoder (2 to 4 Binary Decoder)

- Block diagram: 2 inputs and 4 outputs, plus one enable input.



- **Functional Behaviour** -(the in-out relationship) for binary decoder:
- only an output (1-out-of-2^n) can be asserted when a code is applied to the inputs.

Inputs			Outputs			
en	a[1]	a[0]	y[3]	y[2]	y[1]	y[0]
0	d	d	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

- **Logic Function**

Function:

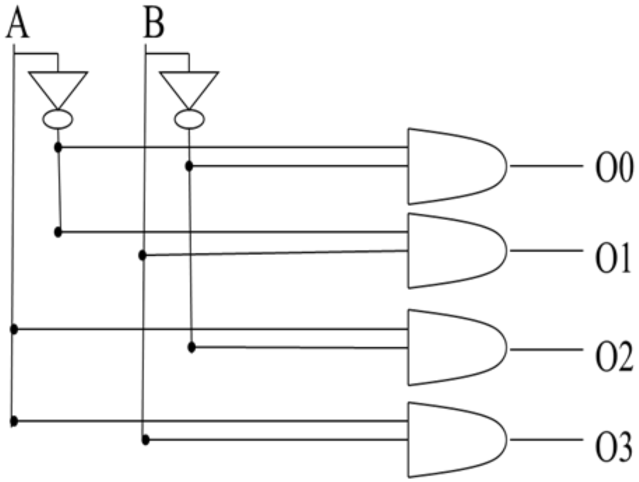
$O_3O_2O_1O_0$ = 0001 when A_1A_0 = 00 → $O0 = \overline{A1} \cdot \overline{A0}$

$O_3O_2O_1O_0$ = 0010 when A_1A_0 = 01 → $O1 = \overline{A1} \cdot A0$

$O_3O_2O_1O_0$ = 0100 when A_1A_0 = 10 → $O2 = A1 \cdot \overline{A0}$

$O_3O_2O_1O_0$ = 1000 when A_1A_0 = 11 → $O3 = A1 \cdot A0$

- Implementation using basic logic gates



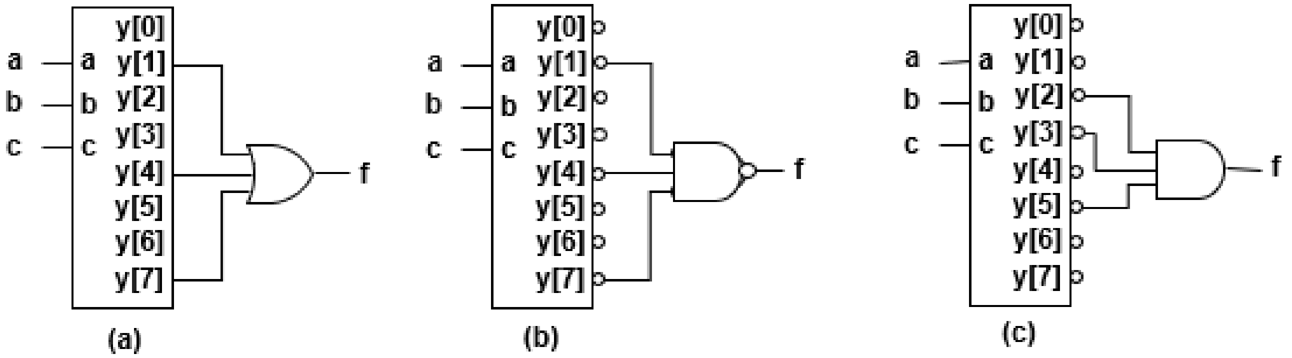
Decoder Application 1: Minterm Generation

- Notice the outputs of binary decoder represent all possible minterms.
  i. Can be used as a minterm generator.

Example10 : Implement the following functions.

1. f(A,B,C) = m1 + m4 +m7: use a decoder (with active-high outputs) with an OR gate.
2. f(A,B,C) = (m1’ m4’ m7’)’: use a decoder (with active-low outputs) with a NAND gate.
3. f(A,B,C) = m2’ m3’ m5’: use a decoder (with active-low outputs) with an AND gate.

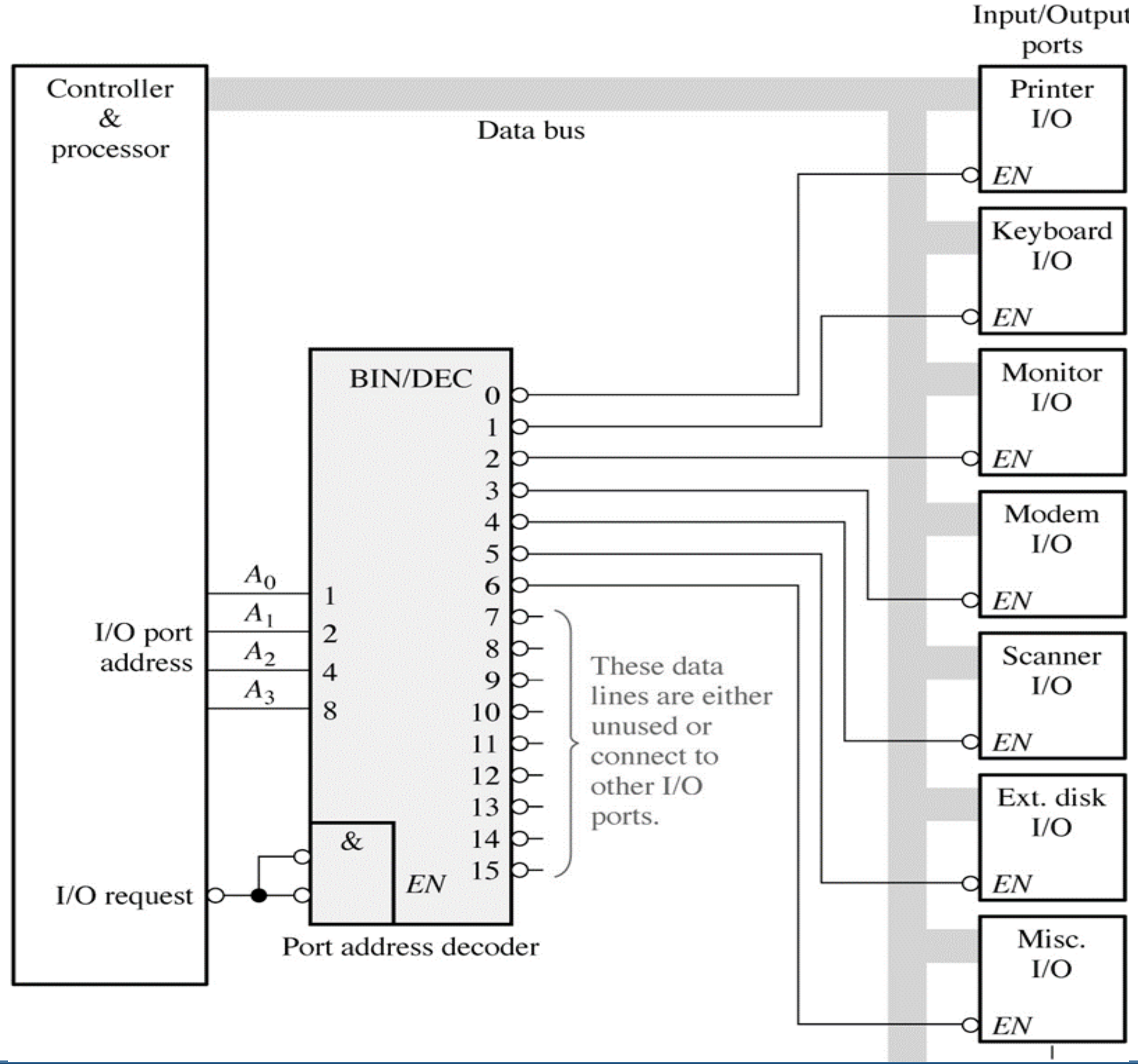
Solution:



Decoder Application 2: Device Enable/Control

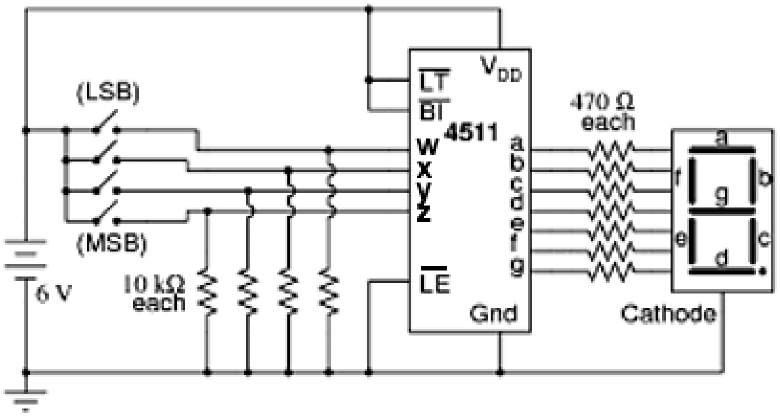
Example 11:

A 4-to-16 binary decoder to selectively enable one out of 7 I/O devices. 9 outputs are not used. Only one device is allowed to be active at any time.

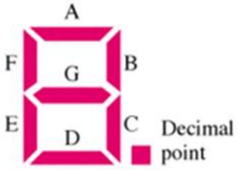


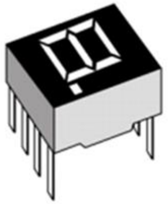
Display decoder

- A display decoder converts a code into a suitable format to drive a numeric display (LCD, 7-segment display, dot matrix display, etc)
- Example
  i. 7-segment display decoders (BCD-to-7 segment decoders)
  ii. Available in the market as standard components 7447 and 74X49.



Inputs	Outputs						
w x y z	a	b	c	d	e	f	g
0 0 0 0	1	1	1	1	1	1	0
0 0 0 1	0	1	1	0	0	0	0
0 0 1 0	1	1	0	1	1	0	1
0 0 1 1	1	1	1	1	0	0	1
0 1 0 0	0	1	1	0	0	1	1
0 1 0 1	1	0	1	1	0	1	1
0 1 1 0	1	0	1	1	1	1	1
0 1 1 1	1	1	1	0	0	1	0
1 0 0 0	1	1	1	1	1	1	1
1 0 0 1	1	1	1	1	0	1	1
1 0 1 0	d	d	d	d	d	d	d
1 0 1 1	d	d	d	d	d	d	d
1 1 0 0	d	d	d	d	d	d	d
1 1 0 1	d	d	d	d	d	d	d
1 1 1 0	d	d	d	d	d	d	d
1 1 1 1	d	d	d	d	d	d	d

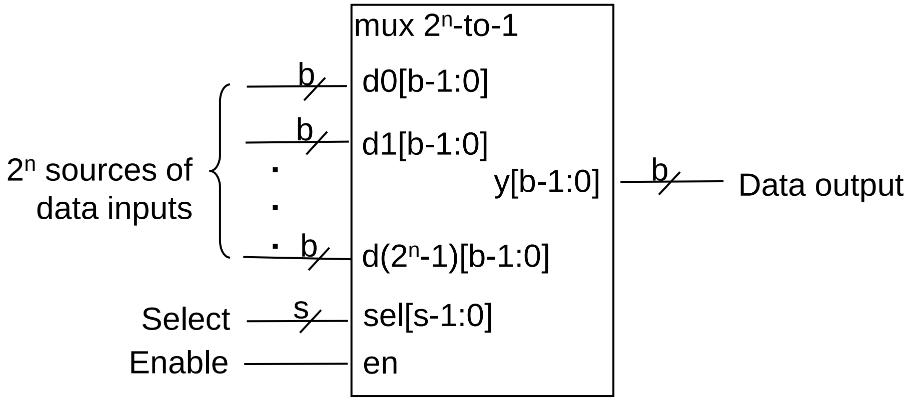




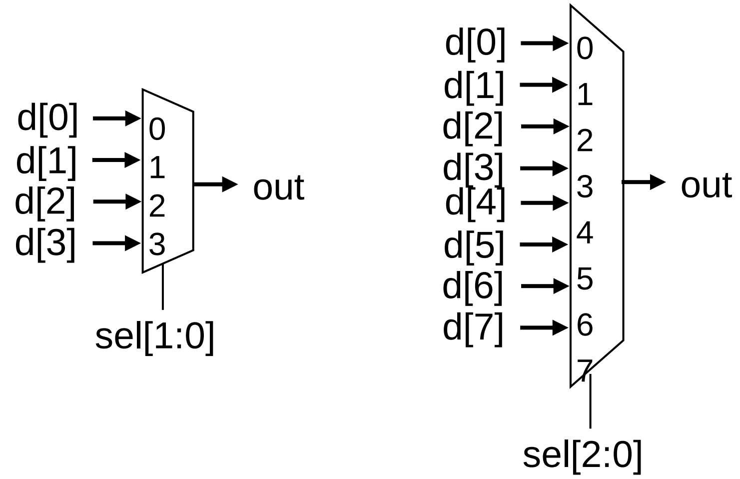
Multiplexer (Data Selector)

- A multiplexer (mux) is a circuit that has:
  i. 2^n data inputs
  ii. s data selectors (control inputs)
  iii. An output

- Like many other circuits, the mux can have additional enable input signal, en
  i. To enable or disable the operation of the mux
  ii. Set the output to z



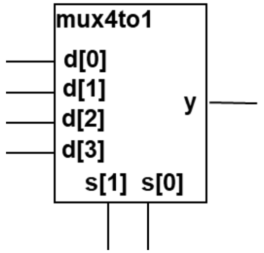
Block diagram of 2^n-to-1 mux



Graphical symbol of 4-to-1 mux and 8-to-1 mux

Multiplexer (4 to1 Multiplexer)

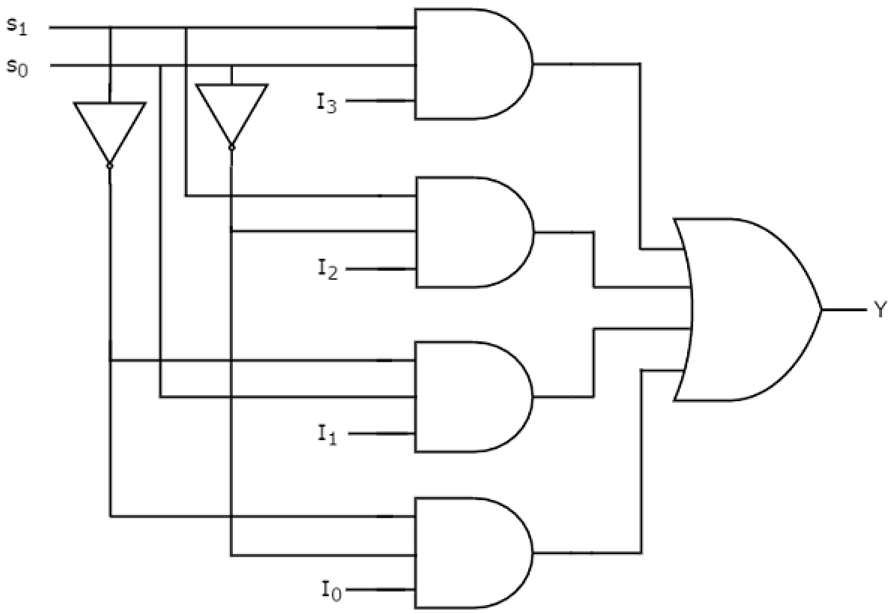
- Block diagram
  - A basic 4-to-1 multiplexer has the following signals.
  - The output signal y.
  - The input signals, d[3:0].
  - Since the multiplexer has 4 data inputs, it required 2 bits data-select ($4 = 2^2$).



- Functional Behaviour

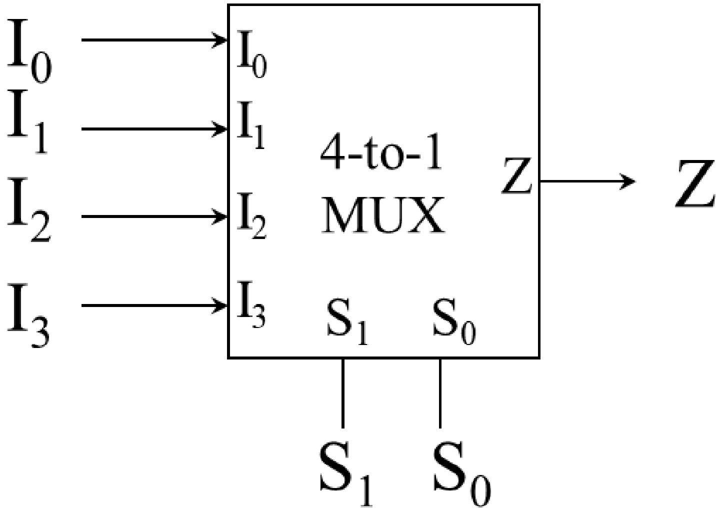
s[1]	s[0]	y
0	0	d[0]
0	1	d[1]
1	0	d[2]
1	1	d[3]

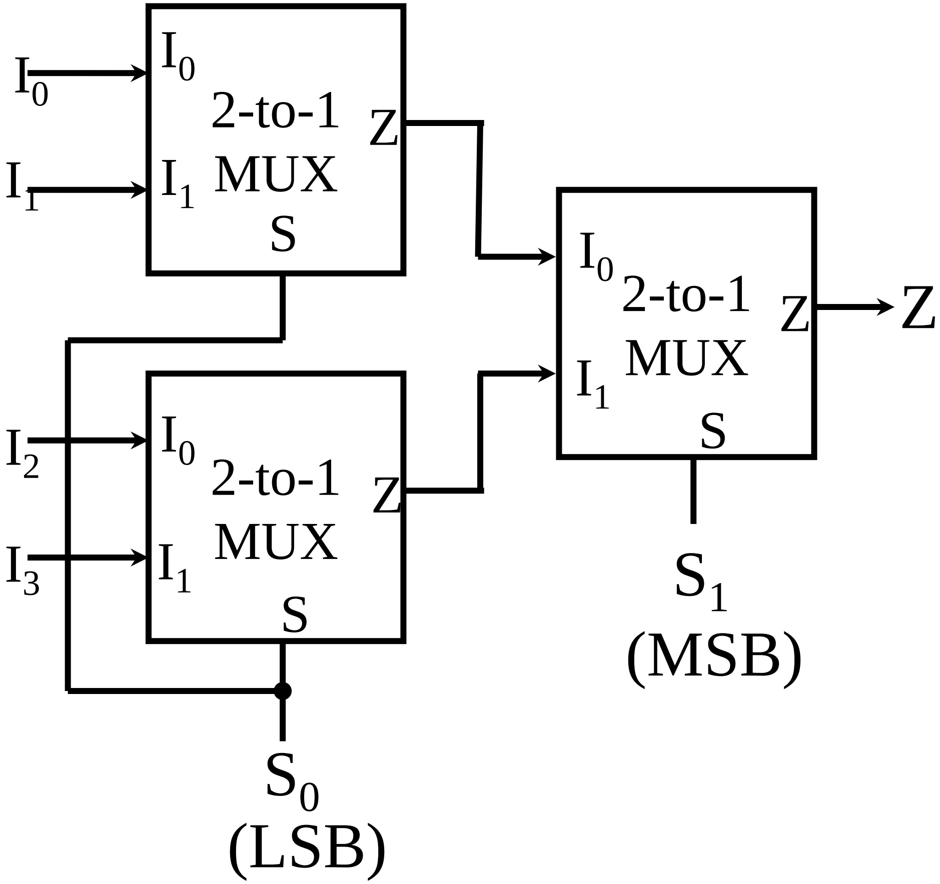
- Logic function
  ✓The logic function of the multiplexer:
  y = s[1]'s[0]'d[0] + s[1]'s[0] d[1] + s[1]s[0]'d[2] + s[1]s[0]d[3]



Multiplexer Expansion

- A few multiplexers can be combined to built a bigger multiplexer.
- For example: a 4-to-1 MUX can be built by combining three units of 2-to-1 MUX are required.





Application: Minterm Generation

- Select inputs -> generates minterms, while data lines are used to enable the minterms

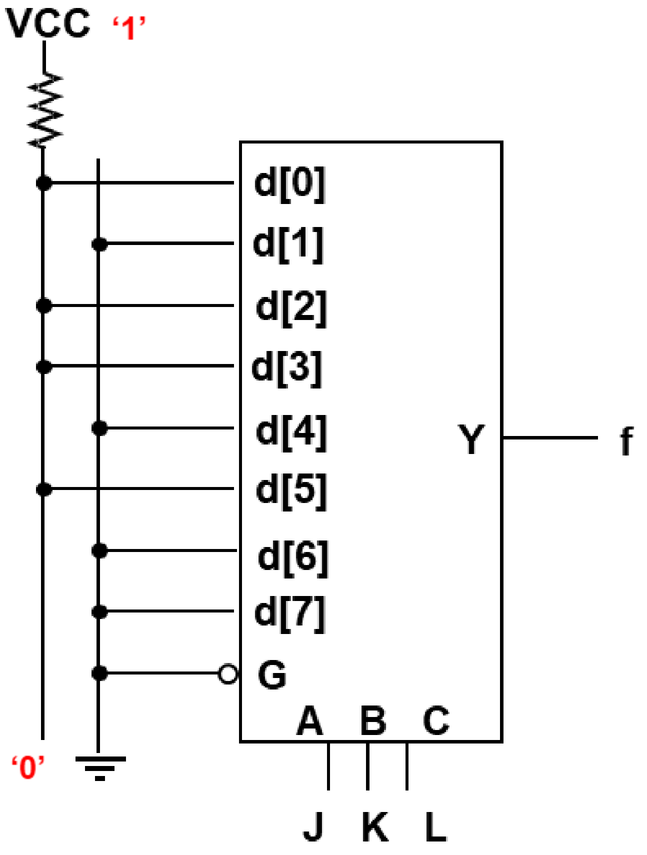
Example 12: Use a 8-to-1 multiplexer to realize $f(J,K,L) = \Sigma m(0, 2, 3, 5)$.

Solution

i. List the truth table for the function.

ii. Then map the content of the truth table to the mux.

J K L	f
0 0 0	1
0 0 1	0
0 1 0	1
0 1 1	1
1 0 0	0
1 0 1	1
1 1 0	0
1 1 1	0



Application: Minterm Generation

Example 13:

i. Implement f(a,b,c) = ab + b'c using 4-to-1 mux if a and b are used as the select signal of the mux.

Solution

Part (a)

- Express the logic function in canonical form.

  f = ab + b'c

  = ab(c' + c) + b'c(a' + a)

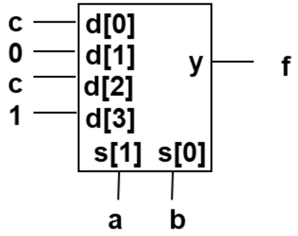
  = abc' + abc + a'b'c + ab'c

- Construct its truth table.

- Derive the reduced inputs (Mux inputs)

a	b	c	f	Mux inputs
0	0	0	0	d[0] = c
0	0	1	1	
0	1	0	0	d[1] = 0
0	1	1	0	
1	0	0	0	d[2] = c
1	0	1	1	
1	1	0	1	d[3] = 1
1	1	1	1	

- The final result:



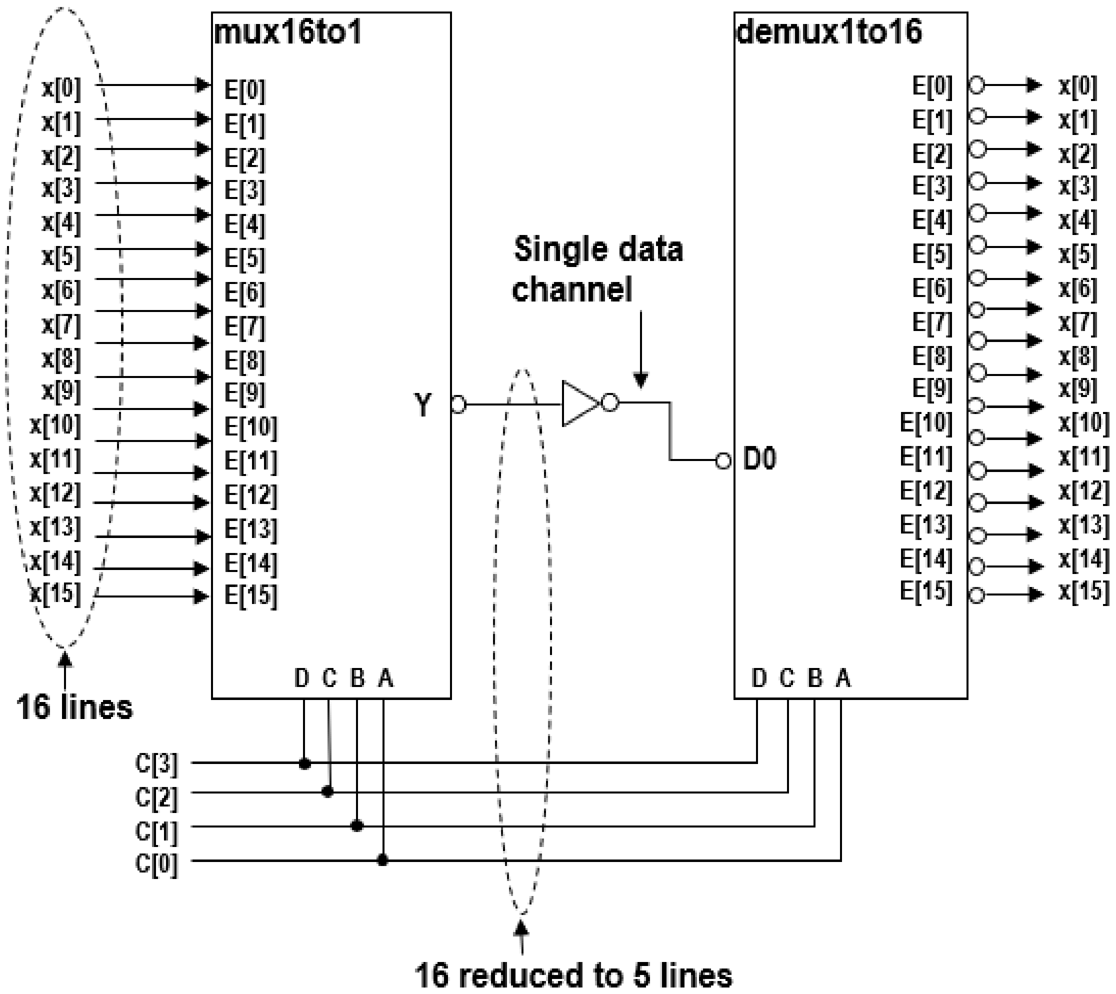
**If not specified, other variables can be used as the select input

Application of Mux/Demux: Data Routing/Transmission*

- Data must be switched from multiple sources to a destination.

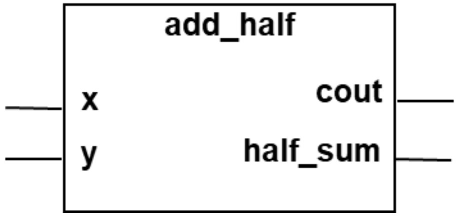
Example 14: Design a 16-line to 16-line mux/demux system.

- The purpose of the system is to reduce the number of wires used for transmission



Half Adder & Full Adder

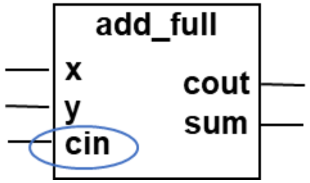
- A half-adder is a logic circuit that performs a single bit addition. It adds two bits, x and y, and produces two outputs, a sum bit and a carry out bit.
- There is no carry-in input
  i. carry-in is treated as 0.
- Block diagram



- Truth table:

x	y	cout	half_sum
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

- The main difference between a full adder and half-adder is that a full adder accepts the carry in C_{in}.
- Block diagram



- Truth table

cin	x	y	cout	sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Adder

- Addition is one of the most common operation in digital systems.
- Adder is an arithmetic circuit that adds two n-bit binary numbers.
- Can be implemented in combinational or sequential logic.
- Terminology for addition operation:

$$\underbrace{11110_2}_{\textbf{Augend}} + \underbrace{01100_2}_{\textbf{Addend}} = \underbrace{01010_2}_{\textbf{Sum}}$$

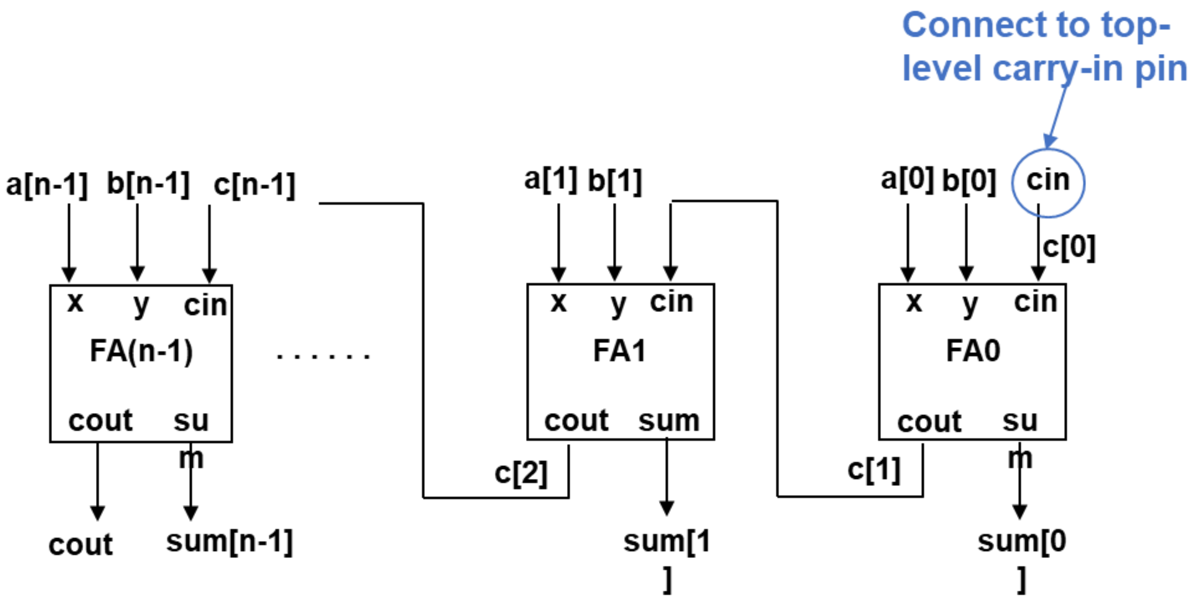
- The carry bit 1 is called the carry output, Cout of the third column and the carry in, Cin to the fourth column of addition.

	1←	1←	1←	0←	0←	
	1	1	1	1	0	Augend
						Carries
+	0	1	1	0	0	Addend
	0	1	0	1	0	Sum

N-bit adder: Carry –Ripple adder

- n-bit carry-ripple adder adds 2 n-bit binary numbers (signed and unsigned)
  - ✓ Result is (n+1)-bit: n-bit sum and a carry-out.
  - ✓ Each bit position requires a full-adder

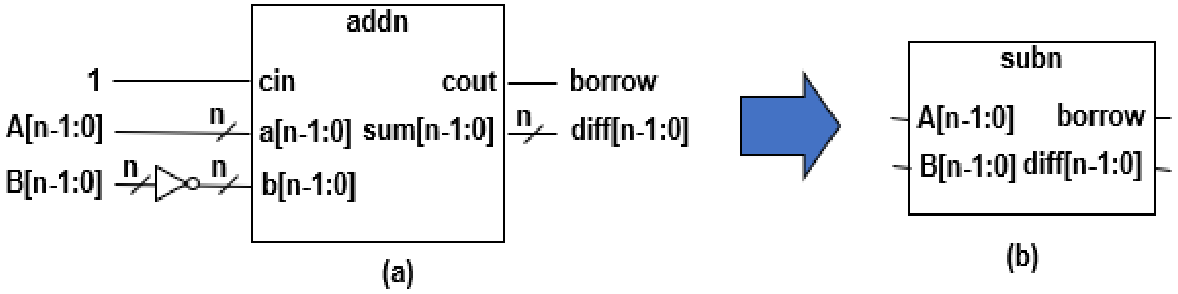
- Principle in constructing an n-bit carry-ripple adder
  - ✓ Cascading n full-adders (FA) through a chain of carry signals.
  - ✓ FA(i).cout => FA(i+1).cin where i = 0, 1, …. n-1



Subtractor Constructed from Adder

- Subtraction can be carried out using the adder circuit
- Based on the 2's complement arithmetic

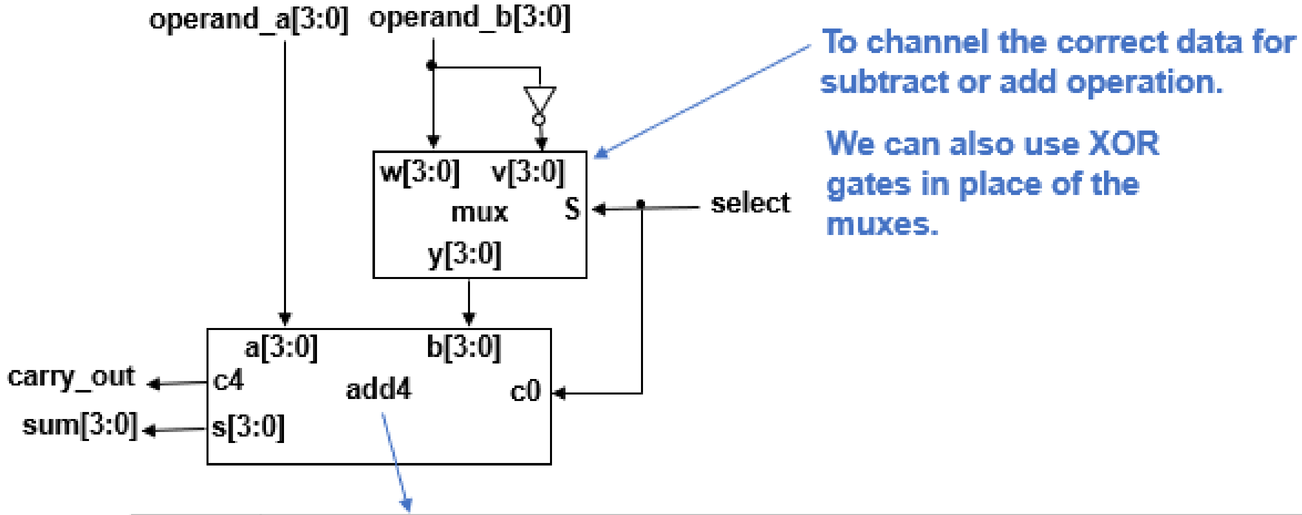
$$
\begin{aligned}
D_2 &= A_2 - B_2 \\
&= A_2 + [B]_2 \\
&= A_2 + (B_2)' + 1
\end{aligned}
$$



- However, the circuit can only perform one function i.e., subtraction.
  - Cannot perform addition.
- To perform both, add and subtract functions, combine the adder and subtracter circuits into one.

Subtraction/Addition Implementation Based on 2's Complement Arithmetic

- Construction of binary adder used to perform both addition and subtraction.



Signals	Addition Operation	Subtraction Operation
select	select = 0 Mux output, y[3:0] = w[3:0] = operand_b[3:0]	select = 1 Mux output, y[3:0] = v[3:0] = operand_b[3:0]'
c0	c0 = select = 0	c0 = select = 1
sum[3:0]	sum[3:0] = a[3:0] + b[3:0] + c0 = operand_a[3:0] + y[3:0] + 0 = operand_a[3:0] + w[3:0] = operand_a[3:0] + operand_b[3:0]	sum[3:0] = a[3:0] + b[3:0] + c0 = operand_a[3:0] + y[3:0] + 1 = operand_a[3:0] + v[3:0] + 1 = operand_a[3:0] + operand_b[3:0]' + 1

Overflow

- Two's complement numbers (2cns) uses complement arithmetic.
  - ✓ E.g. (A – B) can be performed by computing A + (-B) = A + B_{2C}
  - ✓ Only need a binary adder and complementing circuits to handle both add and subtract.
  - ✓ Easy hardware implementation.
- One problem of digital systems is that it have limited bits for data representation (limited range of signed values).
  - ✓ The decimal range for an n-bit digital system using 2cns is $(-2^{n-1}) \leq N \leq (2^{n-1} - 1)$.
  - ✓ E.g. an 8-bit digital system can only operate on values between -128 to 127.
- **If the result is out of the range, signed overflow occurs.**
  - ✓ The **out-of-range result cannot be used.**
  - ✓ Should design digital system that generates a warning signal. So that invalid numbers are not mistaken for correct results.

Overflow: Signed and Unsigned Numbers

- The computation for signed and unsigned overflows are not the same.

  Example

  Compute (– 9 – 5).

  $$-9 - 5 = (-9) + (-5)$$
  $$= [01001] + [00101]$$
  $$= 10111_{2c} + 11011_{2c}$$

Signed addition: input and output numbers treated as signed

Unsigned addition: input and output numbers treated as unsigned

	Two's complement	Signed	Unsigned
	10111_{2c}	(- 9)	23
	$+ 11011_{2c}$	+ (- 5)	+ 27
Carry out →	110010_{2c}	- 14	50

- For unsigned arithmetic, carry out indicates an overflow.
- For signed numbers, the carry out from the sign-bit is not sufficient to determine the overflow.
  - ✓It is meant for extending the adder circuit to higher order bits.
  - ✓However, overflow can be detected by observing the carry into the sign-bit position and the carry out of the sign-bit position. If these two carries are not equal, an overflow has occurred.

Overflow: 2's Complement Arithmetic for Signed Numbers

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- The following 3 case studies are useful for detecting and analysing overflow.
- Assume B and C are positive in a 5-bit system (range represented: -16 to 15).

Case study 1: A = B + C

Compute 12 + 7.

$$\begin{array}{r} 0\ 1\ 1\ 0\ 0_{2c} \\ +\ 0\ 0\ 1\ 1\ 1_{2c} \\ \hline 1\ 0\ 0\ 1\ 1_{2c} \end{array} \qquad \begin{array}{r} 1\ 2 \\ +\ \ \ 7 \\ \hline +\ 1\ 9 \end{array}$$

Result shows an incorrect sign bit (indicates overflow)

- Addition of two positive numbers produce a negative result
- $[10011]_{2c} = -01101_2 = -13$
- -13 is incorrectly interpreted due to overflow
- Sum requires > 5 bits to represent it

Case study 2: A = - B - C

Compute (–12 – 5).

$$\begin{aligned} -12 - 5 &= (-12) + (-5) \\ &= [01100] + [00101] \\ &= 10100_{2c} + 11011_{2c} \end{aligned}$$

$$\begin{array}{r} 1\ 0\ 1\ 0\ 0_{2c} \\ +\ 1\ 1\ 0\ 1\ 1_{2c} \\ \hline 1\ 0\ 1\ 1\ 1\ 1_{2c} \end{array} \qquad \begin{array}{r} (-1\ 2) \\ +\ (-\ \ 5) \\ \hline -\ 1\ \ 7 \end{array}$$

Carry discarded

Result shows an incorrect sign bit (indicates overflow)

- Addition of two negative numbers produce a positive result
- Incorrectly interpreted as 15 due to overflow

Overflow rule for addition:
If two 2's complement numbers with the same sign are added, then overflow occurs if:
- adding two positive numbers give a NEGATIVE result.
- adding two negative numbers give a POSITIVE result.

Overflow: 2's Complement Arithmetic for Signed Numbers

- **Case study 3: A = B - C**
  - The magnitude of (B - C) will always be less than either of the two numbers.
  - Overflow can never occur

Compute 12 - 5.

12 - 5 = 12 + (-5)

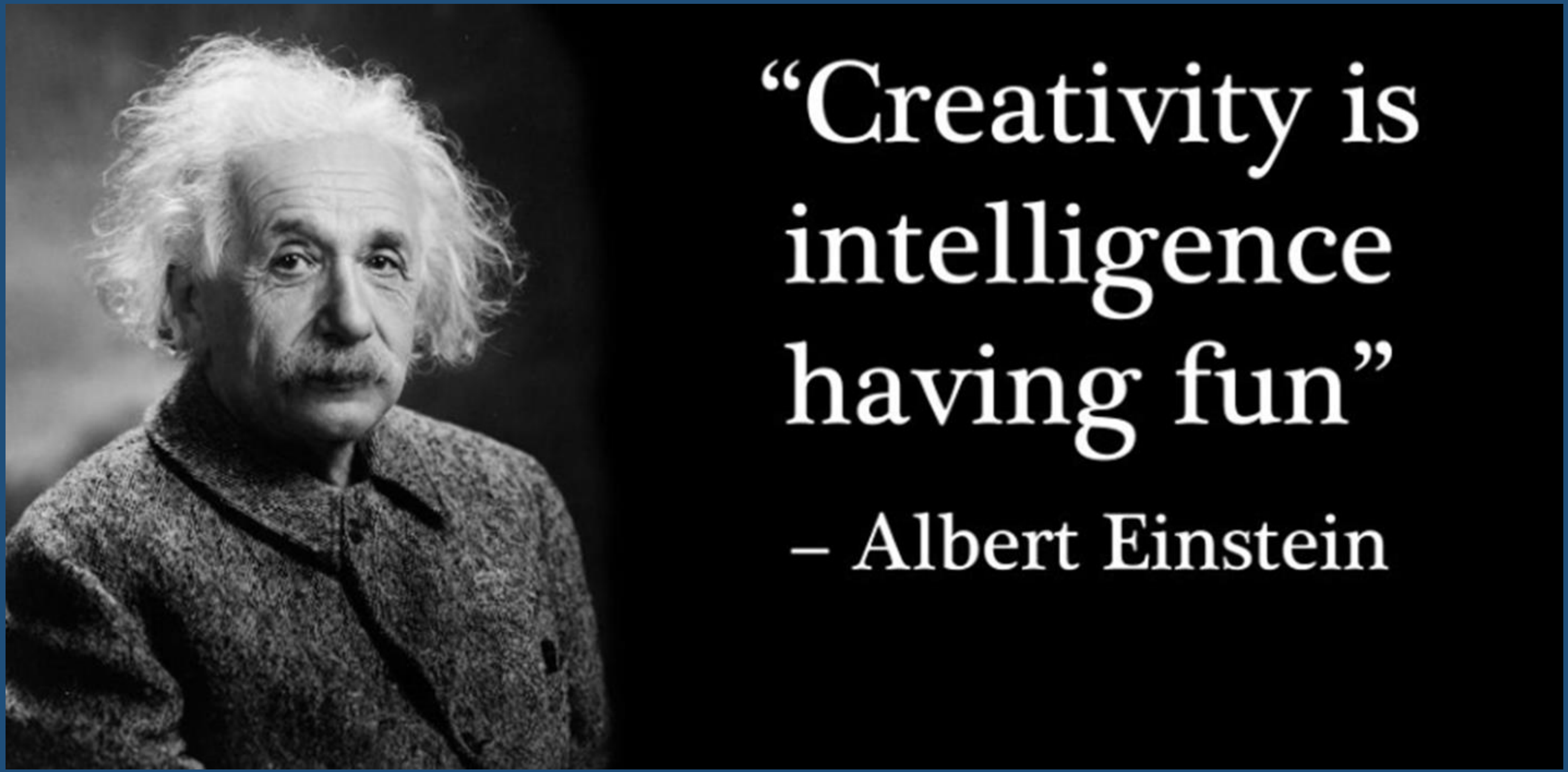
= 01100 + [00101]

= $01100_{2c} + 11011_{2c}$

$$
\begin{array}{r}
0\ 1\ 1\ 0\ 0_{2c} \\
+\ 1\ 1\ 0\ 1\ 1_{2c} \\
\hline
1\ 0\ 0\ 1\ 1\ 1_{2c}
\end{array}
\qquad
\begin{array}{r}
1\ 2 \\
+\ (-5) \\
\hline
+\ \ 7
\end{array}
$$

Carry discarded

The carry generated can be discarded
- It's not a valid carry
  - B > C shouldn't generate a carry



Q & A

Thank you